

Analyze the data and generate insights that could help Netflix decide which type of shows/movies to produce and how to grow the business.

```
import pandas as pd
import numpy as np
```

```
df=pd.read_csv("netflix_titles.csv")
```

```
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	de
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	n
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	A
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	T
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	f
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	ce

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
cat_cols=df.select_dtypes(include=['object']).columns.tolist()
num_cols=df.select_dtypes(include=['int64']).columns.tolist()
```

```
num_cols
```

```
['release_year']
```

```
cat_cols
```

```
['show_id',  
 'type',  
 'title',  
 'director',  
 'cast',  
 'country',  
 'date_added',  
 'rating',  
 'duration',  
 'listed_in',  
 'description']
```

```
df['show_id'].nunique()
```

```
8807
```

```
df.shape
```

```
(8807, 12)
```

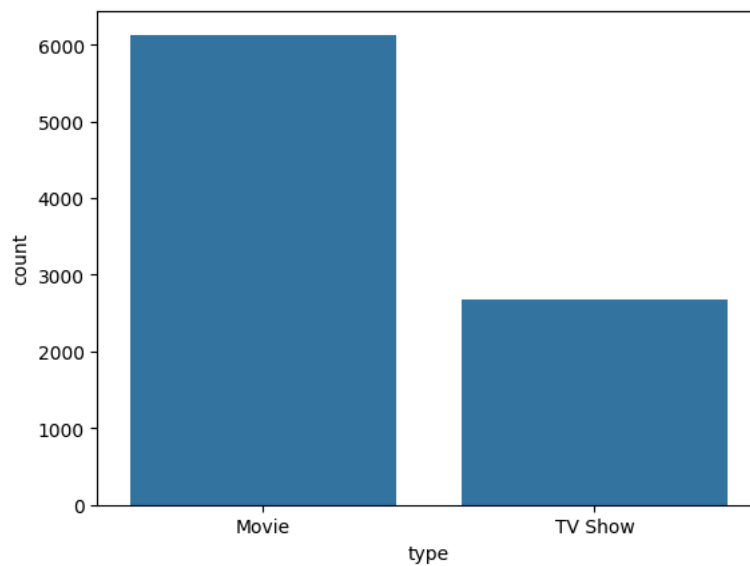
```
df["type"].value_counts()
```

	count
type	
Movie	6131
TV Show	2676

```
dtype: int64
```

```
import seaborn as sns  
sns.countplot(x=df["type"])
```

<Axes: xlabel='type', ylabel='count'>



```
df["director"].unique()
```

```
array(['Kirsten Johnson', nan, 'Julien Leclercq', ..., 'Majid Al Ansari',  
      'Peter Hewitt', 'Mozes Singh'], dtype=object)
```

```
df["director"].value_counts()
```

	count
director	
Rajiv Chilaka	19
Raúl Campos, Jan Suter	18
Suhas Kadav	16
Marcus Raboy	16
Jay Karas	14
...	...
James Brown	1
Ivona Juka	1
Mu Chu	1
Chandra Prakash Dwivedi	1
Majid Al Ansari	1

4528 rows × 1 columns

dtype: int64

df["country"].value_counts()

	count
country	
United States	2818
India	972
United Kingdom	419
Japan	245
South Korea	199
...	...
Mexico, United States, Spain, Colombia	1
Canada, Norway	1
Finland, Germany, Belgium	1
Argentina, United States, Mexico	1
United Kingdom, United States, Germany, Denmark, Belgium, Japan	1

748 rows × 1 columns

dtype: int64

df[["date_added","release_year"]]

	date_added	release_year
0	September 25, 2021	2020
1	September 24, 2021	2021
2	September 24, 2021	2021
3	September 24, 2021	2021
4	September 24, 2021	2021
...	...	...
8802	November 20, 2019	2007
8803	July 1, 2019	2018
8804	November 1, 2019	2009
8805	January 11, 2020	2006
8806	March 2, 2019	2015

8807 rows × 2 columns

df["date_added"]=pd.to_datetime(df["date_added"])

```
df[df['date_added'].dt.year < df["release_year"]]
```

```
df[df['date_added'].dt.year > df["release_year"]]
```

	show_id	type		title	director	cast	country	date_added	release_year	rating	duration	liste
0	s1	Movie		Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documen
7	s8	Movie		Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	1993	TV-MA	125 min	Dr: Indepe M Intern M
16	s17	Movie		Europe's Most Dangerous Man: Otto Skorzeny in ...	Pedro de Echave García, Pablo Azorín Williams	NaN	NaN	2021-09-22	2020	TV-MA	67 min	Document Intern M
17	s18	TV Show		Falsa identidad	NaN	Luis Ernesto Franco, Camila Sodi, Sergio Goyri...	Mexico	2021-09-22	2020	TV-MA	2 Seasons	Crir S Spi Langua Shows,
21	s22	TV Show		Resurrection: Ertugrul	NaN	Engin Altan Düzyatan, Serdar Gökhan, Hülya Dar...	Turkey	2021-09-22	2018	TV-14	5 Seasons	Intern TV Show Ac Advent
...	...	...		...	...	...	...	...	...	...	...	...
8802	s8803	Movie		Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult M Dr: Th
8803	s8804	TV Show		Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kic Kore Show Corr
8804	s8805	Movie		Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Com Horror M
8805	s8806	Movie		Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Chilc Family M Corr
8806	s8807	Movie		Zubaan	Moze Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dr: Intern Movies, & Mu

5542 rows x 12 columns

```
df["rating"].value_counts()
```

show_id	count	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
TV-MA	3207	TV Show	Hilda	NaN	Bella Ramsey, Ameerah Falzon-Ojo, Oliver Nelso...	United Kingdom, Canada, United States	2020-12-14	2021	TV-Y7	2 Seasons	Kid
TV-14	2160										
TV-PG	863										
R	799				Emily Tennant, Shannon Chan-Kent, Kazumi Evans...	Canada, United States, Ireland	2020-11-15	2021	TV-Y	2 Seasons	Kid
PG-13	490	TV Show	Polly Pocket	NaN							
TV-Y7	334										
TV-Y	307										
PG	287	TV Show	Love Is Blind	NaN	Nick Lachey, Vanessa Lachey	United States	2020-02-13	2021	TV-MA	1 Season	Realit Rorr TV S
TV-G	220										
NR	80										
G	41				Candace Cameron Bure, Jodie Sweetin, Andrea Ba...	United States	2019-12-06	2020	TV-PG	5 Seasons	Com
TV-14-FV	3169	TV Show	Fuller House	NaN							
NC-17	3										
UR	3										
74 min	1	TV Show	Maradona in Mexico	NaN	Diego Armando Maradona	Argentina, United States, Mexico	2019-11-13	2020	TV-MA	1 Season	Docu Spa Lang TV S
84 min	1										
66 min	1				Will Arnett, Aaron						
dtype: int64		TV Show	Ro. Jack Horseman			United States	2020-10-25	2020	TV-MA	6 Seasons	Com
df["duration"].value_counts()											
duration	count	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
1 Season	1793	TV Show	The Hook Up Plan	NaN	Marc Ruchmann, Zita Hanrot, Sabrina Ouazani, J...	France	2019-10-11	2020	TV-MA	2 Seasons	Internal TV Sh Rorr TV Sh
2 Seasons	425										
3 Seasons	199										
90 min	152	TV Show	Unbreakable Kimmy Schmidt	NaN	Ellie Kemper, Jane Krakowski, Tituss Burgess, ...	United States	2018-05-30	2019	TV-14	4 Seasons	Com
97 min	146										
...	...										
228 min	1	TV Show	Arrested Development	NaN	Jason Bateman, Portia de Rossi, Will Arnett, M...	United States	2018-05-29	2019	TV-MA	5 Seasons	Com
18 min	1										
205 min	1										
201 min	1										
191 min	1	movie	Hans Teeuwen: Real Rancour	Doesjka van Hoogdalem	Hans Teeuwen	Netherlands	2017-07-01	2018	TV-MA	86 min	Stan Co
220 rows x 1 columns											
dtype: int64					Tuppence Middleton, ...						Crim Show
5658	5659	TV Show	Sense8	NaN	Smith	United States	2016-12-23	2018	TV-MA	2 Seasons	Drama
df.rename(columns={"listed_in":"genre"}, inplace=True)											
df											
5677	5678	TV Show	Tokyo Trial	NaN	Paul Freeman, Irrfan Khan, Sta...	Canada, Japan, Netherlands	2016-12-13	2017	TV-MA	1 Season	Internal TV Sh TV Dr
7063	7064	Movie	Incoming	Eric Zaragosa	Scott Adkins, Aaron McCusker, Vahldin Prelic, ...	Serbia, United States	2018-10-26	2019	TV-MA	89 min	Act Adver Sc Fa
7112	7113	TV Show	Jack Taylor	Stuart	Iain Glen, Killian Scott, ...	United States	2013-03-31	2016	TV-MA	1 Season	Crim Sh Internal

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	TV_Score
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentar
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shows, Dramas, Myster
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajjila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime Shov Internatio TV Shows, Ac
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseri Reality
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	Internatio TV Shov Romantic Shows, TV
...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	2019-11-20	2007	R	158 min	Cult Movie Dram Thrill
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	2019-07-01	2018	TV-Y7	2 Seasons	Kids' Korean Shows, Comed
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	2019-11-01	2009	R	88 min	Comedi Horror Mov
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	2020-01-11	2006	PG	88 min	Childrer Family Movi Comed
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	TV-14	111 min	Dram Internatio Movies, Mu & Music

8807 rows x 12 columns

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df["genre"].str.split(",")
```

```

genre
0 [Documentaries]
1 [International TV Shows, TV Dramas, TV Myste...
2 [Crime TV Shows, International TV Shows, TV ...
3 [Docuseries, Reality TV]
4 [International TV Shows, Romantic TV Shows, ...
...
8802 [Cult Movies, Dramas, Thrillers]
8803 [Kids' TV, Korean TV Shows, TV Comedies]
8804 [Comedies, Horror Movies]
8805 [Children & Family Movies, Comedies]
8806 [Dramas, International Movies, Music & Music...

```

8807 rows x 1 columns

dtype: object

```

all_genre=[]
for rows in df["genre"].str.split(","):
    all_genre.extend(rows)

all_genre_unique=list(set(all_genre))

```

len(all_genre_unique)

73

```
top_5_popular_genre=pd.Series(all_genre).value_counts().head()
```

top_5_popular_genre

	count
International Movies	2624
Dramas	1600
Comedies	1210
Action & Adventure	859
Documentaries	829

dtype: int64

```

movies=df[df["type"]== "Movie"]
shows=df[df["type"]== "TV Show"]

```

movies.shape,shows.shape

((6131, 12), (2676, 12))

```
movies["director"].value_counts()
```